SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE CODE: CSA15

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) Question
Count: 5

Code of Conduct
Total Marks: 25

I _____S.Naveen karthigeyan/192521061_____
(Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: _____

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on

premise computing.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

CO1- AT-1

① Define cloud computing and explain four characteristics

Sol. Cloud computing means using computing services like storage service and software through the internet instead of maintaining them on our own computers.

1. On Demand Self-Services

Users can access services whenever they need them.

e.g. Creating a virtual machine AWS

2. Broad network access

Cloud services can be accessed through mobiles, laptops and tablets using internet.

3. Resource Pooling

Cloud service provide share computing resources among many users.

4. Rapid elasticity

Resource can be included or decreased to demand.

e.g. An online shopping website adding servers.

2. Explain the evolution of cloud computing

Ans.

Cloud computing developed through different stages

Hardware Evolution:

Early computers were large and expensive

Mainframe computing:

Many users shared one powerful computers

Personal computers:

Computers became smallers and affordable.

client-server computing.

client received computer across from servers.

Internet evolution.

The internet connected computer across the world.

Web software:

Applications became accessible through web browsers

3. Explain Server virtualization and differentiate it from cloud computing

Ans:

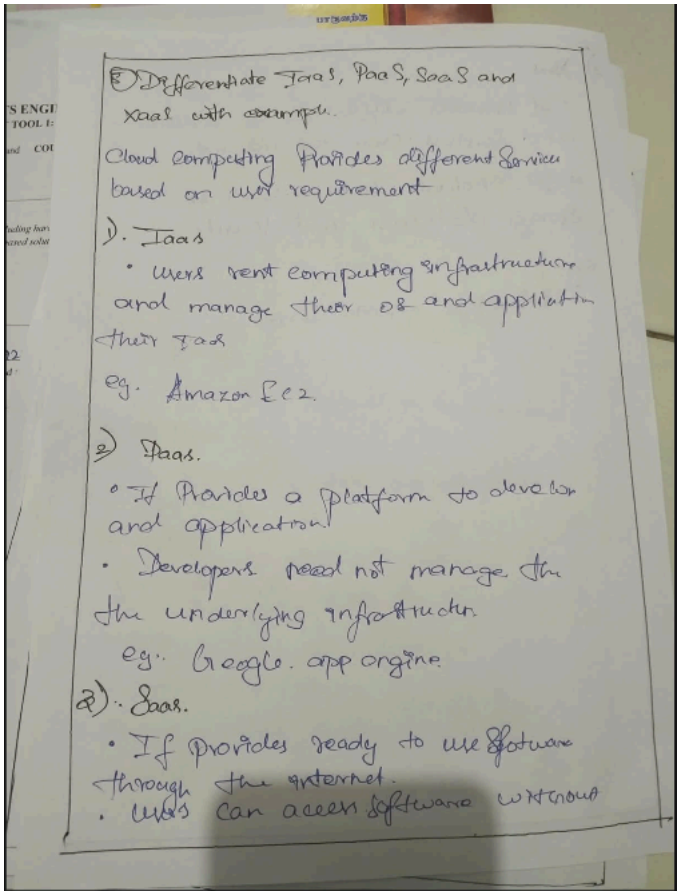
Server virtualization is a technology that gives one physical server into multiple machines each virtual gets own operating system.

Role of Virtualization

=> It improve hardware virtualized.
=> It reduces the need for servers
=> It allows multiple os to run on one server.
=> It helps allocate resource easily

Differences

Virtualization	cloud computing
=> It is a technology	=> It is a service model
=> It creates virtual machines	=> It provides computing services
eg: multiple vms on	=> eg: Hosting Server



5) Differentiate IaaS, PaaS, SaaS and XaaS with example.

Cloud computing Provides different Services based on user requirement

1). IaaS

- Users rent computing infrastructure and manage their OS and application their IaaS

eg. Amazon EC2.

2) PaaS.

- It Provides a platform to develop and application.
- Developers need not manage the the underlying infrastructure.

eg. Google. app engine

3). SaaS.

- It provides ready to use software through the internet.
- Users can access software without

4) Xaas.

- It Provides different IT resources and services over the internet.
- It provides services such as storage database and security

eg:
Microsoft azure server